Shape sensitivities of a DSMC flow

The ensemble adjoint and the seven terms of the shape gradient

2026-10-09

Outline

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Notation: step k, time step Δt, F_{num} molecules per simulator particle, superscript m = the state after the move of step k (before its collisions).

The kinetic problem

Flow domain $\Omega(\theta) = \text{box} \setminus B(\theta)$; the body is a C^1 level set

$$B = \{x : \phi(x; c) < 0\}, \qquad \phi(x; c) = \sum_j c_j B_j(x), \qquad n = \nabla\phi/|\nabla\phi|,$$

B_j cubic B-splines on knots of spacing h_ϕ, n pointing into the flow. Boltzmann equation for the one-particle density $f(t, x, v)$ in $\Omega(\theta)$:

$$\partial_t f + v \cdot \nabla_x f = Q(f, f),$$

inflow (drifting Maxwellian flux) at $x = 0$, outflow at $x = L$, specular y walls (or, in free space, the inward flux on every face), and on the body, for $v \cdot n > 0$,

$$|v{\cdot}n|\, f(x, v) = \int_{v'\cdot n<0} K(v' \to v;\ n)\, |v'{\cdot}n|\, f(x, v')\, \mathrm{d}v', \quad K = (1-a)\,\delta(v - R_n v') + a\, q_w(v \mid n),$$

$R_n v' = v' - 2(v' \cdot n)\, n$; q_w the diffuse half-Maxwellian of the wall. A coefficient change moves the surface along its normal by $V_n = -\delta\phi/|\nabla\phi| = -\sum_j \delta c_j B_j/|\nabla\phi|$: every control is a normal displacement, chained through $c(\theta)$.

The objective: windowed drag

Mean x-force on the body over a window of W steps:

$$D(\theta) = \frac{1}{W\,\Delta t} \int_{t_1}^{t_2} \int_{\partial B} \int_{v\cdot n<0} m \left(v - v^{\mathrm{out}}(v)\right)_x |v\cdot n|\, f \,\mathrm{d}v \,\mathrm{d}S \,\mathrm{d}t.$$

In DSMC it is the expectation of a sum of *per-hit rewards*:

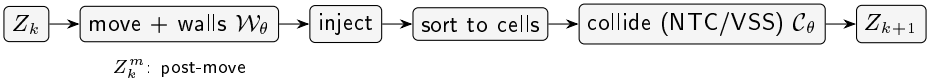
$$D = \mathbb{E}\Big[\sum_{k\in \text{window}} \; \sum_{\text{body hits } j \text{ of step } k} r_{k,j} \Big], \qquad r_{k,j} = \frac{F_{\mathrm{num}}}{W\,\Delta t}\, m \left(v^{\mathrm{in}}_{k,j} - v^{\mathrm{out}}_{k,j}\right)_x .$$

For a diffuse hit v^{out} is the drawn re-emission velocity.

Squash study: $J(s) = D(s) - \lambda_A\, A(s)$, an ellipse with semi-axes R and sR, $A(s) = \pi R^2 s$ exact, λ_A fixed so that the free-molecular optimum is $s^* = 0.35$. Only $\mathrm{d}D/\mathrm{d}s$ needs the adjoint. No state reward, no terminal reward.

DSMC as a Markov chain

State $Z_k = \{(x_i, v_i, a_i)\}_{i=1}^{N}$ over fixed slots (a = alive). One step:



Common random numbers. Every stage draws from a key fixed by (seed, step, stage), so with the key fixed each stage is a deterministic function of its input: a run can be recorded and replayed exactly, and two geometries see the same random numbers.

Where θ enters the forward

(a) the move: hit times, hit points, normals, specular $v^{\mathrm{out}} = R_n v^{\mathrm{in}}$, bounce chains;

(b) the per-hit reward itself, through $v^{\mathrm{out}}(n)$;

(c) the diffuse re-emission law $q_w(\cdot \mid n)$;

(d) the hit set: which particles hit, and in which step;

(e) the collisions: NTC's attempt count carries the cut-cell flow volume $V_c(\theta)$.

NTC collisions and the cut-cell volume

In cell c with N_c particles and flow volume V_c, the attempt count is a stochastic rounding of

$$\bar{n}_{\mathrm{att},c} = \tfrac{1}{2}\, N_c (N_c - 1)\, \Sigma_c\, \Delta t\, F_{\mathrm{num}} / V_c,$$

each attempt draws a pair uniformly and accepts it with probability $q(g)/\Sigma_c$, $q(g) = \sigma_T(g)\, g$ the VSS rate, Σ_c a majorant; an accepted pair scatters with the VSS law. Hence

$$\Pr\{\text{pair } (i,j) \text{ collides in step } k\} \approx \frac{q(g_{ij})\, \Delta t\, F_{\mathrm{num}}}{V_c} \quad \propto \quad \frac{1}{V_c(\theta)},$$

and a particle's expected collisions per step are $\nu \Delta t \approx (2\bar{n}_{\mathrm{att},c}/N_c)\, \mathbb{E}[q/\Sigma_c]$.

$V_c(\theta) = |c \cap \Omega(\theta)|$: a cut cell's collision probabilities depend on the shape. A cell that the body splits into disconnected flow regions gets one collision cell per region.

$\mathrm{Kn} = \lambda_{\mathrm{mfp}}/2R = 0.5$ on the squash case: $\nu \Delta t \approx 2.5\,\%$ per step; free-molecular: 0.

The value function

Value of one particle at the post-move instant of step k:

$$\lambda_k(x,v) = \frac{\delta D}{\delta f_k^m(x,v)} = \mathbb{E}\big[\text{rewards collected from } t_k^m \text{ on } \big| \text{ one particle at } (x,v)\big].$$

For a Boltzmann gas it solves the adjoint equation, backward in time,

$$-\partial_t \lambda - v\cdot\nabla_x\lambda = \mathcal{L}_f^*\lambda + r, \qquad (\mathcal{L}_f^*\psi)(v) = \int f(w)\int q\left[\psi(v')+\psi(w')-\psi(v)-\psi(w)\right]\mathrm{d}\sigma\,\mathrm{d}w,$$

with, at a specular wall, $\lambda(x_w, v^{\mathrm{in}}) = r(v^{\mathrm{in}}) + \lambda(x_w, R_n v^{\mathrm{in}})$, and at a diffuse wall the expectation over the re-emission.

Particle-carried form. The PDE is never solved. One value per particle, at that particle's own state, is carried backward through the recorded run:

$$\mu_{k+1} \xrightarrow{\text{collision pull-back}} \lambda_k^m \xrightarrow{\text{move pull-back}} \mu_k,$$

μ_k the values at the start of step k, λ_k^m at its post-move state.

Backward through the move

A particle that survives the move of step k keeps its value and collects the rewards of the hits it made during the move:

$$\mu_{k,i} = \mathbf{1}\left[a^m_{k,i} \wedge \neg\,\mathrm{born}_{k,i}\right]\left(\lambda^m_{k,i} + \sum_j r_{k,j,i}\right).$$

- Specular walls are deterministic maps: the value moves with the particle.
- A diffuse wall is a Markov kernel; its realised re-emission is a sample of it, so the one carried value estimates $\mathbb{E}[\lambda(\text{after}) \mid \text{before}]$ without bias (tower property), however many bounces the step has.
- Particles absorbed or leaving the box carry 0; a particle born in step k carries nothing to t_k (it did not exist).

Backward through the collisions: virtual partners

The collision step applies $e^{\Delta t\,\mathcal{L}^*}$ to the values; it is applied in S substeps. In each, every particle i of cell c draws K partners j_κ uniformly from the other particles of c, and fresh VSS angles:

$$\lambda_i \mathrel{+}= \sum_{\kappa=1}^{K} w_{i\kappa}\big[\Lambda(v_i') + \Lambda(v_{j_\kappa}') - \Lambda(v_i) - \Lambda(v_{j_\kappa})\big], \qquad w_{i\kappa} = \frac{q(g_{i\kappa})}{\Sigma_c}\,\frac{2\,n_{\mathrm{att},c}}{N_c}\,\frac{1}{KS},$$

Λ one smooth fit of the current values in the cell (next frames). The weight is the expected number of collisions of i in the step (attempts per particle $\times$ acceptance) spread over KS samples; all four values come from one fit, so their errors cancel.

- Substep s fits on the velocities as they stood at that point of the NTC sweep ("staged" nodes, reconstructed from the record's slot order); a particle whose velocity changed across a substep boundary moves its value through the fit.
- Splitting error $O(\Delta t/S)$; noise far below one realised pair per collision.

The collision increment of slot i in step k, $c_{k,i} = \sum_{\text{substeps}} \sum_\kappa w_{i\kappa}\Phi_{i\kappa} \approx \Delta t\,(\mathcal{L}^*\lambda)_i$, is kept: the along-flight, order and cut-volume terms use it.

The value field of the collision pull-back: one polynomial per cell

The pull-back needs $(3+2K)N$ field values per substep (the node, two ghosts per virtual partner, the boundary move). In the scaled coordinates $\tilde{p} = (x/\Delta x,\ v/v_{\mathrm{s}})$, $v_{\mathrm{s}} = \sqrt{kT/m}$, each collision cell c gets **one** least-squares fit of its nodes' carried values, at offsets $o = \tilde{p} - \bar{p}_c$ from the node mean:

$$\phi(o) = \big(1,\ o_x,\ o_v,\ \{o_{v,a}o_{v,b}\}_{a\le b},\ o_x^2\big), \qquad \Big(\sum_{n\in c}\phi_n\phi_n^{\top} + \rho\,\bar{d}\,\mathrm{diag}(0,1,\dots,1)\Big)\beta_c = \sum_{n\in c}\phi_n\lambda_n,$$

and every query in c reads $\Lambda(x_q, v_q) = \phi(\tilde{p}_q - \bar{p}_c)\cdot\beta_c$.

- **Order guard**: only the columns of the orders whose cumulative count is $\le n_c/1.5$ (14 columns in 2-D need 21 nodes; else a plane, else the cell mean).
- **Next to a surface** (λ jumps across $v\cdot n = 0$): three fits per cell – nodes leaving, approaching ($v\cdot n_c \gtrless 0$, n_c the cell's mean nearest-surface normal) and all; a query takes its side's fit, the unsplit one when no surface is near, its side is empty, or the cell's normals disagree ($|\sum n_i|/\#n_i < 0.9$: corners, opposed faces).
- Exact for any field quadratic in v (the per-cell fit reproduces its own monomials).

Why one fit per cell, and why the guard

- **The collision term sees only the non-invariant part of λ**: for $\lambda \in \mathrm{span}(1, v, |v|^2)$ every four-term $\Lambda(v_i') + \Lambda(v_j') - \Lambda(v_i) - \Lambda(v_j)$ vanishes. What the pull-back needs is the cell population's v-structure beyond the invariants – a property of the cell, not of a neighbourhood of each query.
- The per-query fit over the k nearest nodes depended on k ($k = 32 \to 64$: $+1.6 \pm 0.7\,\%$ of $\mathrm{d}D/\mathrm{d}s$, 12 seeds; flat from $k = 64$, where it spans the whole cell and its weights only range over $[\frac{1}{2}, 1]$) and cost $\propto k$. The per-cell fit has no k: C fits instead of $19N$.
- A cell's values are mostly noise (the cell mean predicts held-out nodes at the noise floor). An *underdetermined* quadratic in a sparse cell answers the partner ghosts with extrapolated noise, and the backward recursion compounds it: the unguarded per-cell quadratic is unstable (one seed in 12: $+97\,\%$). The guard bounds the columns by the nodes.

Measured (Kn 0.5 squash vs CRN FD): at production density (16 seeds) guarded per-cell $-3.7 \pm 0.8\,\%$, per-query $k = 32$ $-3.4 \pm 0.8\,\%$, a cubic-in-v guarded fit $-0.7 \pm 0.8\,\%$; a seed $1100 \to 46$ s. Stable on the optimiser's complicated shapes, where the per-query one-sided fit diverged.

The per-query fit: the shape gradient's slopes

The shape terms need the slopes $\nabla_x\lambda$, $\nabla_v\lambda$ at reflected states, with node sets chosen per query (next frame). For a query (x_q, v_q): the k nearest admissible nodes in $\tilde{p}$, weights $\omega_n = 1/\big(1 + (d_n/d_k)^2\big)$, the same quadratic design at $\delta = \tilde{p}_n - \tilde{p}_q$:

$$\Big(\sum_n \omega_n \phi_n \phi_n^\top + \rho\,\bar{d}\,\mathrm{diag}(0, 1, \ldots, 1)\Big)\beta = \sum_n \omega_n \phi_n \lambda_n \qquad (\bar{d}\text{: mean diagonal; } \rho = 10^{-8}),$$

$$\Lambda = \beta_0, \qquad \nabla_x\Lambda = \beta_{1:1+D}/\Delta x, \qquad \nabla_v\Lambda = \beta_{1+D:4+D}/v_{\mathrm{s}}.$$

- Pooled over the 3^D block ($k = 96$): the slopes need more nodes than a value.
- The slopes are the weak link: the value field is discontinuous next to a body (next frame).

Which nodes: functions of the current state only

Next to a body the post-move population splits: particles about to hit carry their imminent hit's reward, particles that just left do not. λ jumps across that boundary, and a polynomial fitted across it has a wrong slope. The shape-gradient fits therefore use, for a query whose last hit had normal n at x_q:

1. **one-sided**: only nodes leaving that surface, $v \cdot n > 0$;
2. **pooled**: the $k = 96$ nearest such nodes from the 3^D block of cells around the query's cell (variance);
3. **half-space**: only nodes with $(x_{\mathrm{node}} - x_q) \cdot n \geq -\frac{1}{4}\Delta x$ (a body thinner than the block would otherwise put its far side in the pool).

Rule: stratify by the current state, never by the future. With $\lambda = P\lambda_1 + (1-P)\lambda_0$ (P the probability of some future event),

$$\nabla\lambda = P\nabla\lambda_1 + (1-P)\nabla\lambda_0 + \nabla P\,(\lambda_1 - \lambda_0);$$

a fit inside classes defined by the future event estimates the first two terms only. Classes defined by the present state ($v \cdot n$, position) keep all three.

The first variation of one step

Write one step as $Z' = G_\theta(Z, \omega)$, randomness $\omega \sim p_\theta$, $R_\theta(Z, \omega)$ the rewards paid at the wall during the move, $V(Z')$ the value of what follows. Then

$$\frac{\mathrm{d}}{\mathrm{d}\theta}\mathbb{E}\big[R + V(Z')\big] = \underbrace{\mathbb{E}\big[\nabla V \cdot \partial_\theta G\big]}_{\text{pathwise (frozen } \omega)} + \underbrace{\mathbb{E}\big[\partial_\theta R_\theta\big]}_{\text{hit reward (frozen } \omega)} + \underbrace{\mathbb{E}\big[(R+V)\,\partial_\theta \log p_\theta(\omega)\big]}_{\text{score}}$$
$$+ \underbrace{\int_{\Sigma_\theta} [\![R+V]\!]\, u_\Sigma \,\mathrm{d}\sigma}_{\text{boundary}} + \underbrace{\mathbb{E}\big[\partial_\theta(\text{collision law) at fixed particles}\big]}_{\text{operator}},$$

Σ_θ: where G_θ jumps (hit $\to$ miss; hit into the next step), $[\![\cdot]\!]$ the jump, u_Σ its speed in θ. Summed over steps with $V = \lambda$, this gives the seven terms:

pathwise:	1 transport of the perturbed state (with the along-flight identity)
hit reward, score:	2 hit reward (tally); 3 diffuse normal
boundary:	4 hit $\leftrightarrow$ miss; 5 window edges; 6 hit/collision order
operator:	7 cut-cell volume

The replayed wall map

Under the fixed keys the move of step k is a pure function of the geometry:

$$(x_k^{\mathrm{post}},\ v_k^{\mathrm{post}},\ T_k,\ \mathcal{E}_k) = \mathcal{W}_\theta(x_k, v_k;\ \kappa_k),$$

T_k surface tallies, $\mathcal{E}_k$ the traced events (x^h, v^{in}, v^{out}, n, t^h per hit). It is re-run from the record and differentiated by reverse-mode AD:

- hit time: $\phi(x + tv; c) = 0$, so $\partial t^h / \partial c_j = -B_j(x^h)/(v \cdot \nabla\phi(x^h))$ (bisection for the root, one Newton step that carries the derivative);
- then $x^h = x + t^h v$, $n(x^h; c)$, $v^{\mathrm{out}} = R_n v^{\mathrm{in}}$, $x^{\mathrm{post}} = x^h + v^{\mathrm{out}}(\Delta t - t^h)$, repeated for every bounce of the step;
- grazing hits: $1/(v \cdot \nabla\phi)$ is floored at $\epsilon\,|v|\,|\nabla\phi|$ in the derivative only (the primal hit is exact): bounded per-hit weights, bias $O(\epsilon)$.

The replay must reproduce the record: any flipped discrete outcome is an error.

Term 1: transport of the perturbed state (pathwise)

$$\text{pathwise}_k = \sum_{i \in \text{surv}} \nabla_x \lambda_k^m(x_i^{\text{post}}, v_i^{\text{post}}) \cdot \partial_\theta x_i^{\text{post}} + \nabla_v \lambda_k^m(x_i^{\text{post}}, v_i^{\text{post}}) \cdot \partial_\theta v_i^{\text{post}}.$$

- $\partial_\theta x^{\text{post}}$, $\partial_\theta v^{\text{post}}$: the VJP of the wall map; nonzero only for particles that hit the body in step k, so the slopes are fitted at those states only.
- $\nabla_v \lambda$: reflected velocities. On a specular body $\partial_\theta v^{\text{out}}$ is $O(1)$ per hit; the slope says how the particle's future depends on where it goes.
- $\nabla_x \lambda$: shifted hit points and hit times. Its component along the particle's own velocity comes from an identity, not from the fit (next frame).
- Only the survivors of the step contribute ($a_k \wedge a_k^m \wedge \neg\, \text{born}_k$).

For a convex body in free-molecular flow a reflected particle never re-hits, so this term is the blockage (flow-field) feedback; with collisions it carries the gas's response to the deflected particles.

$\nabla_x \lambda$ along the flight: the along-flight identity

Free flight carries $(x - sv, v)$ at time t to (x, v) at $t + s$. So

$$\lambda_t(x - sv, v) = \lambda_{t+s}(x, v) + s\,(\mathcal{L}^*\lambda)(x, v) + O(s^2), \qquad \lambda_{t+s}(x, v) = \lambda_t(x, v) + s\,T(x, v) + O(s^2),$$

$T =$ the change of the objective per unit delay of the particle's whole future (the hits that cross the window edges), and $s\,\mathcal{L}^*\lambda =$ the collisions during the extra time. Hence

$$v \cdot \nabla_x \lambda = -T - \mathcal{L}^*\lambda.$$

- T is the value of the *timing rewards* (Term 5's band rewards), carried by a second value column; $\mathcal{L}^*\lambda \approx c_{k,i}/\Delta t$ from the collision pull-back.
- The fitted $\nabla_x \lambda$ keeps its perpendicular part; its component along v is replaced:

$$\nabla_x \lambda \leftarrow \nabla_x \lambda - \frac{v \cdot \nabla_x \lambda + T + c_k/\Delta t}{|v|^2}\, v.$$

- Accuracy: NTC gives a particle one collision chance per step wherever it sits in its cell, so in the discrete scheme the $\mathcal{L}^*$ part holds to $O(\nu \Delta t)$; exact without collisions.

Term 2: the hit reward (tally)

The value λ_k^m starts *after* the move, so the rewards the move itself pays — the body hits of step k — enter the backward by hand, $\mu_k = \lambda_k^m + \sum_j r_{k,j}$; their derivative is this term, the value's is Term 1. The reward $r = \frac{F_{\mathrm{num}}}{W\Delta t}\, m(v^{\mathrm{in}} - v^{\mathrm{out}})_x$ depends on θ through $v^{\mathrm{out}} = R_n v^{\mathrm{in}}$ (and through v^{in} of later bounces in the same step). The surface tallies are outputs of the wall map, so this is one more cotangent:

$$\mathrm{hit}_k = \frac{\partial D}{\partial T_k} \cdot \partial_\theta T_k, \qquad \frac{\partial D}{\partial T_k} = \frac{F_{\mathrm{num}}}{W\Delta t}\, \mathbf{1}[k \in \text{window}] \text{ on the body's } x\text{-momentum.}$$

Exact given the hits; it needs no fitted field.

Example: specular cylinder of radius r, free-molecular. With impact parameter b, $F \propto \int_{-r}^{r} \cos^2\varphi \,\mathrm{d}b = 4r/3$; differentiating at fixed rays gives $\int 2b^2/r^3 \,\mathrm{d}b = 4/3$, so this term alone is F/r — the whole derivative of an isolated body, since the new grazing rays carry $\cos^2\varphi = 0$. The other terms carry the response of the flow (blockage, collisions).

Term 3: diffuse walls, the normal of the re-emission law

A diffuse hit re-emits from $q_w(v \mid n) = \frac{v_n}{s^2} e^{-v_n^2/2s^2} \, \frac{e^{-|v_t-u|^2/2s^2}}{2\pi s^2}$, $s^2 = kT_w/m$. Its normal moves with θ. Split the hit's future into its reward and the value after it:

$$\frac{\mathrm{d}}{\mathrm{d}n}\mathbb{E}\big[r(v^{\mathrm{out}}) + V(v^{\mathrm{out}})\big] = \underbrace{\mathbb{E}\Big[\frac{\partial r}{\partial v^{\mathrm{out}}} \cdot \frac{\partial v^{\mathrm{out}}}{\partial n}\Big]}_{\text{pathwise, reparameterised}} + \underbrace{\mathbb{E}\big[V(v^{\mathrm{out}})\, \partial_n \log q_w(v^{\mathrm{out}} \mid n)\big]}_{\text{score, value after the hit}}.$$

- Pathwise part: $v^{\mathrm{out}} = s\,(\rho\, n + z_1 t_1(n) + z_2 t_2(n)) + u$ at fixed draws; the tangent frame is parallel-transported from the primal one (bounded derivative), and the reward enters through the tally (Term 2).
- Score part: weighted by the value-to-go *after* the hit; no fitted slope. $1/v_n$ is floored at $\epsilon|v^{\mathrm{out}}|$ (bias $\approx \epsilon/2$ of the v_t/v_n part).
- Specular bodies: zero.

Term 4: hit ↔ miss (silhouette band)

Along a particle's undeflected line, $\phi_{\min} = \min_s \phi(x + s\hat{v})$: < 0 it hits, > 0 it misses. A surface displacement flips the outcome where $|\phi_{\min}| < |\delta\phi|$:

$$\left.\frac{\mathrm{d}D}{\mathrm{d}\theta}\right|_{\mathrm{sil}} = -\mathbb{E}\left[\rho(0)\,(V_{\mathrm{hit}} - V_{\mathrm{miss}})\,\frac{\partial\phi_{\min}}{\partial\theta}\right] \approx -\frac{1}{2\varepsilon_b}\sum_{|\phi_{\min,i}|<\varepsilon_b}(V_{\mathrm{hit},i} - V_{\mathrm{miss},i})\,B_j(x_i^*)\,\frac{\partial c_j}{\partial\theta},$$

$\partial\phi_{\min}/\partial c_j = B_j(x^*)$ at the closest point (envelope theorem).

- Realised outcome: its reward-to-go. Virtual outcome: the fitted value at the virtual post-move state plus the virtual hit's reward, valued from the leaving population.
- Each passing counted once per trajectory (a band wider than one move would otherwise count it several times).
- Specular: $V_{\mathrm{hit}} \to V_{\mathrm{miss}}$ at grazing incidence, the term is $O(\varepsilon_b)$ and is left out. Diffuse: $O(1)$ — about half of a diffuse body's derivative.

Term 5: hits that cross a window edge

The VJP differentiates a fixed assignment of hits to steps. A surface displacement shifts hit times, and a hit at the end of step k can slip into step $k+1$. For the per-hit drag this changes the objective only at the window's edges:

$$\Delta J_{k,j} = +r_{k,j} \text{ (last window step)}, \qquad -r_{k,j} \text{ (step before the window)}.$$

Boundary term: (density of hit times at the step end) $\times(-\Delta J)$, the density estimated over the hits of the w whole steps ending at each edge:

$$\text{cotangent on } t^h_{k,j}: \qquad -\frac{\pm r_{k,j}/w}{\Delta t} \quad \text{for every body hit of those steps.}$$

- Zero in expectation on a steady window, but part of the windowed objective's derivative; it is included. $w = 10$ steps: $\sim$50$\times$ the hits of a fraction-of-a-step band, at an $O(w\Delta t)$ smoothing of the edge's hit-time density.
- The same per-hit band rewards are the *timing rewards* whose value T enters the along-flight identity.

Term 6: the order of a hit and the step's collisions

NTC collides once per step, after the move. A body hit at the end of step k that slips into step $k+1$ is, during step k's collisions, the *incoming* particle (x_w, v^{in}) instead of the *reflected* one (x_w, v^{out}). Both futures coincide after the next move, so the objective changes by

$$J_c = c(x_w, v^{\mathrm{in}}) - c(x_w, v^{\mathrm{out}}), \qquad c = \Delta t\, \mathcal{L}^* \lambda,$$

at *every* step boundary, not only at the window's edges. The hit times of a steady flow are uniform within the step, so the density at the step end is $1/\Delta t$ and

$$\text{cotangent on } t^h: \qquad +\frac{J_c}{\Delta t} \quad \text{for every body hit.}$$

- c at the hit point: fitted from the collision increments $c_{k,i}$ of the hit's cell, on the approaching side ($v \cdot n < 0$) for v^{in} and the leaving side for v^{out}.
- Without collisions $c \equiv 0$; then Term 5 is the whole hit-step dependence.

Term 7: the cut-cell volume in the collision step

Every collision probability of cell c scales as $1/V_c$ (NTC frame). The collision law is the only thing that changes, so this is a kernel-parameter (operator) term with $\partial_\theta \log q$ replaced by $\partial_\theta \log V_c^{-1}$:

$$\left.\frac{\mathrm{d}D}{\mathrm{d}\theta}\right|_V = -\sum_k \sum_c \frac{\partial_\theta V_c}{V_c}\, C_{k,c}, \qquad C_{k,c} = \tfrac{1}{2}\sum_{i\in c} c_{k,i} \quad \text{(each pair seen from both ends)},$$

$C_{k,c}$ the expected change of the objective by the collisions of cell c in step k. The geometry factor is the shape derivative of the cut volume:

$$V_c = \int_c H(\phi)\,\mathrm{d}x, \qquad \frac{\partial V_c}{\partial c_j} = \int_c \delta(\phi)\, B_j\,\mathrm{d}x = \int_{\Gamma\cap c} \frac{B_j}{|\nabla\phi|}\,\mathrm{d}s,$$

evaluated with a smoothed Heaviside $H_w(\phi) = \frac{1}{2}\big(1+\tanh(\phi/w)\big)$; each sub-region of a split cell separately. Zero without collisions.

Assembly

Per seed, on the level-set coefficients:

$$\widehat{\nabla_c D} = \sum_k \Big(\underbrace{\text{pathwise}_x + \text{pathwise}_v}_{1} + \underbrace{\text{tally}}_{2} + \underbrace{\text{score}}_{3} + \underbrace{\text{silhouette}}_{4} + \underbrace{\text{timing}}_{5} + \underbrace{\text{order}}_{6} \Big) + \underbrace{\text{volume}}_{7},$$

then chained to the controls by the VJP of $\theta \mapsto c(\theta)$, and averaged over seeds.

term	free-molecular, specular	free-molecular, diffuse	collisional
1 pathwise x, v	yes	yes	yes ($+\mathcal{L}^*\lambda$ along v)
2 tally	yes	yes	yes
3 diffuse normal score	0	yes	if diffuse
4 silhouette	$O(\varepsilon_b)$, off	yes	if diffuse
5 window edges	yes	yes	yes
6 hit/collision order	0	0	yes
7 cut-cell volume	0	0	yes

One gradient evaluation: forward with capture → replay → two-column value backward (with the collision increments) → one VJP scan over the captured steps → volume term → control chain.

From the gradient to the optimiser

Squash (one scalar control). Adam on $J'(s) = \widehat{\mathrm{d}D/\mathrm{d}s} - \lambda_A \, \mathrm{d}A/\mathrm{d}s$, the step's largest change of s set by the learning rate; S common-random-number seeds per iterate.

Free shape (all coefficients), at fixed area.

1. seed mean g; restrict to the coefficients within a band of the surface and smooth with a Gaussian (an H^1-type gradient that removes knot-scale noise);
2. gradient descent or Adam (Adam's normalised step is smoothed again);
3. project the gradient and the step onto the area's tangent space, $g - \nu \nabla A$ with $\nu = g \cdot \nabla A / |\nabla A|^2$ (Adam's moments track the constrained gradient);
4. fixed largest normal displacement per iterate ($s_{\max} = 0.05$, a tenth of a cell);
5. reinitialise ϕ to a signed distance;
6. restore the area by a uniform shift γ of ϕ: only the step's $O(s_{\max}^2)$ remainder (sub-mm), so the constraint holds exactly (a retraction);
7. keep the body a margin from every box face: ϕ at the knots floored at margin $- \, d_{\mathsf{face}}$, alternated with the shift until both hold.

What is exact, and what is approximate

piece	approximation	size / control
wall-map VJP, tally, scores	exact given the record	—
collision pull-back	S substeps, K virtual partners	$O(\Delta t/S)$; $(S,K)=(4,8)$
value field of the pull-back	one guarded quadratic per cell (split at surfaces)	$-3.7 \pm 0.8\,\%$ (Kn 0.5)
fitted slopes $\nabla\lambda$	local quadratic, state-based node sets	fit bias; k, pool, half-space
along-flight $\mathcal{L}^*$ part	continuous identity in a discrete scheme	$O(\nu\Delta t)$
grazing floor	$1/(v\cdot n)$ floored in the derivative	$O(\epsilon)$; $\epsilon = 0.05$
silhouette band	finite band density	$O(\varepsilon_b)$
window-edge band	w whole steps	$O(w\Delta t)$ smoothing
order term	uniform hit times in the step	steady flow
captured horizon	sensitivities older than the captured steps	box memory (walls)

Left out on purpose: hits by particles born in the step or leaving the box in it (monitored, 0 while the body is away from the inflow and outflow faces); the change of cell membership of a hit particle ($O(\nu\Delta t)$ below the order term); the silhouette term of specular bodies.

Validation of each piece against finite differences: the validation deck.